Question 1 **10 marks**

Given that the position of an object, as a function of time, in the x direction, is described by the graph below:

(a) What is the *speed* of the object, in the x direction at $t = 0.5$ s?

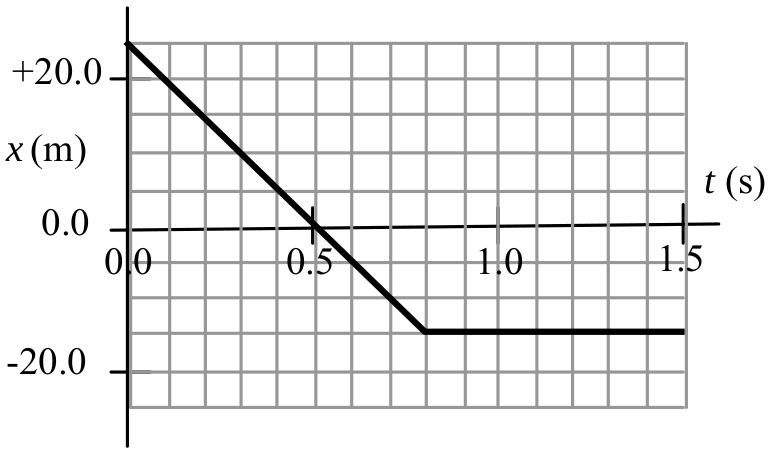
(b) What is the *velocity* of the object, in the x direction at $t = 0.5$ s?

(c) Draw the corresponding velocity time graph.

(d) In what direction is the object travelling at $t = 1.0$ s?

(e) What is the displacement of the object between $t = 0$ s and $t = 1.5$ s?

(f) What is the average velocity of the of the object in the from $t = 0$ s to $t = 1.5$ s?



[2 + 1 + 3 + 1 + 1 + 2 = 10 marks]

Question 2 **15 marks**

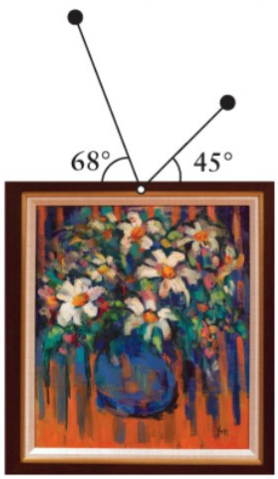
(a) If an object is moving at constant velocity, what can you say about the net force acting on that object?

(b) Consider an example where you are standing on the Earth and holding an apple in your hand. For such an example state two action reaction pairs [both of which must include the apple].

(c) Consider a cricket ball and a tennis ball (each with the same mass). If you drop the tennis ball and cricket ball from the same height (you may ignore air resistance), explain why it hurts more when the cricket ball lands on your foot as compared to the tennis ball.

(d) Describe in words what kinetic (μ_k) and static (μ_s) friction means.

(e) A picture frame is hung using two cords on a wall, as illustrated in the figure to the right. If the mass of the frame is *350* g:



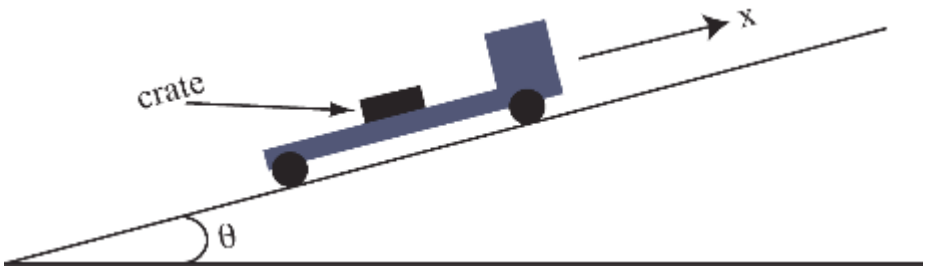
(i) Calculate the tension in each of the cords?

(ii) If the left string is cut what is the tension in the right string?

[2 + 2 + 3 + 2 + (4 + 2) = 15 marks]

Question 3 **9 marks**

Consider a flatbed truck, carrying a crate up an incline, as illustrated in the figure below.



(a) Draw the force diagram for the crate, when the flatbed truck is stationary, labelling each force.

(b) Show that the total force acting on the crate, along the x-direction, is

$$F_x = -\,\mathrm{gm}\sin(\theta) + f_s$$

where m is the mass of the crate, g is acceleration due to gravity and f_s is the static frictional force between the crate and the flatbed truck.

(c) Assuming that the coefficient of static friction is 0.35 and the angle of the incline is 10 degrees calculate the maximum acceleration of the truck before the crate begins to slide.

[3 + 2 + 4 = 9 marks]

Question 4 **11 marks**

Please provide *brief* answers to the following questions. Approximately one paragraph for each part is sufficient. Diagrams may be used to support your explanations.

a) According to Newton's third law, when you push on a shopping trolley in the supermarket, the trolley pushes back on you with an equal and opposite force. Explain why it is that you can cause the trolley to move by pushing it, even though these forces are equal and opposite. Your explanation should refer to Newton's laws and all of the forces involved.

b) A communications satellite occupies a circular geostationary orbit about the earth (that is, it remains fixed above the same point on the earth's equator as the earth rotates). In this orbit, does the satellite experience a force? If so, what is it?

c) In the sport of Sumo Wrestling, two athletes are placed inside a ring drawn on a floor. To win, one must push the other outside of the ring. Why is it an advantage to be very heavy in this sport?

[4 + 3 + 4 = 11 marks]

Question 5 **8 marks**

Please provide *brief* answers to the following questions. Approximately one paragraph for each part is sufficient.

(a) Imagine you are an astronaut standing on the moon and you simultaneously drop a hammer and feather? Which hits the surface of the moon first? The Hammer, the feather or do they both hit at the same time? Explain your reasoning.

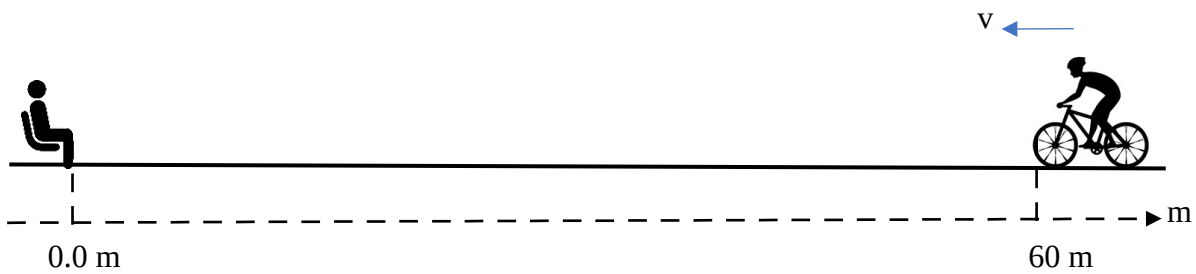
(b) Now imagine you're standing on the Earth and drop the hammer and feather at the same time. Which hits the surface of the Earth first? The Hammer, the feather or do they both hit at the same time? Explain your reasoning.

(c) A book is placed on a desk. A second book is placed on top of it doubling the force due to gravity on the desk. What happens to the atoms of the desk that enables the normal force from the desk to provide exactly the force equal and opposite the two books to prevent any acceleration?

[2 + 3 + 3 = 8 marks]

Question 6 **12 marks**

You're sitting on a bench watching a cyclist riding directly towards you. You are equipped with a radar gun which enables you to accurately measure the distance of the cyclist and his velocity. When he is 60 m distant, you observe he is riding towards you at $-4.0\ ms^{-1}$. Then 2.0 seconds later you measure his velocity as $-7.0\ ms^{-1}$.



(a) Calculate his acceleration over this 2.0 s period.

(b) Assuming his acceleration is constant, calculate how far away from you is he after this 2.0 s.

(c) The cyclist then travels towards you at the constant velocity of $-7.0\,ms^{-1}$. Calculate how far away from you he is after a further 3.0 seconds.

(d) The cyclist then sees you and puts on his brakes to stop before hitting you. Calculate the acceleration required to bring him to rest before hitting you.

[2 + 4 + 2 + 4 = 12 marks]

Question 7 **20 marks**

Tegan Philip is the goal attack of the 2017 Melbourne Vixens's netball team. We shall analyze how Tegan shoots a goal from a distance 4.800 metres to the ring. This situation is shown in figure 1 below.

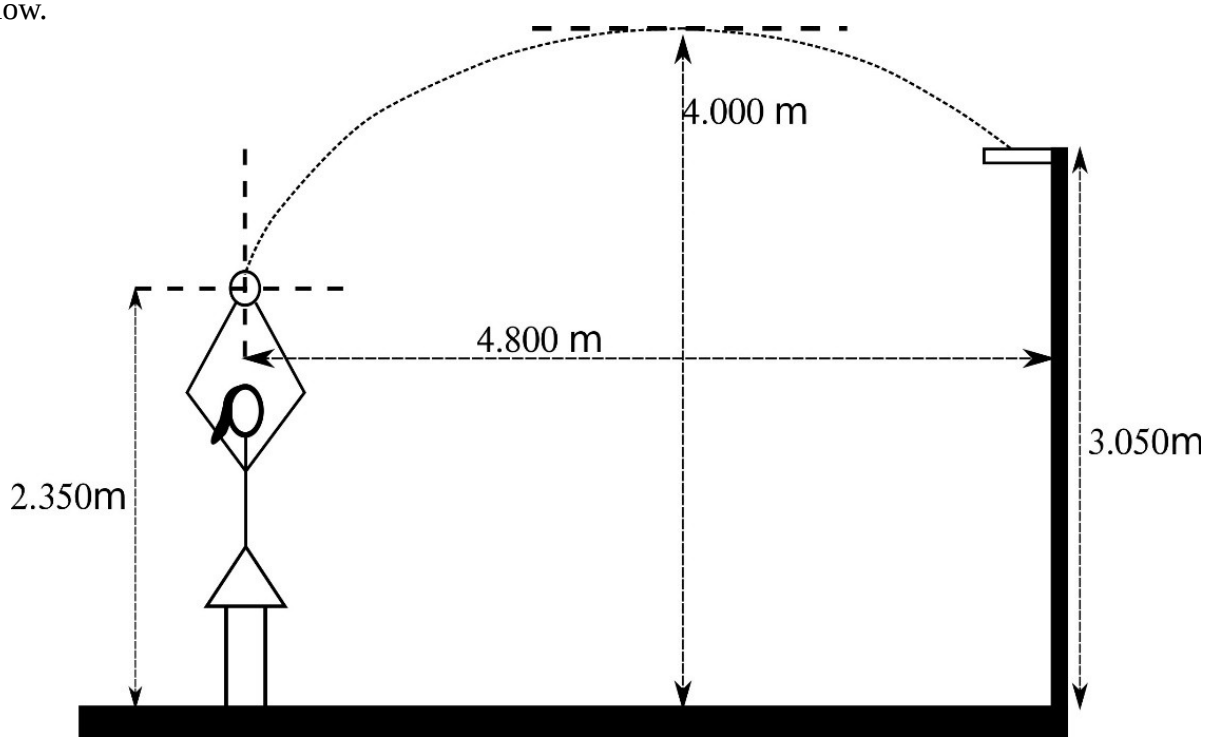


Figure 1. Tegan Philip shoots a goal 4.800 m from the ring. The netball ring is 3.050 m from the ground. The maximum height of the ball during its flight is 4.000 m. Tegan stands still, throws and releases the ball from a height of 2.350 m from the ground. Assume the acceleration due to gravity is 9.800 $m\,s^{-2}$.

(a) Write down the equations for the location of the ball in the horizontal x and vertical y dimensions as a function of time t. Use expressions including the initial horizontal velocity of ball v_{ix}, the initial vertical velocity of the ball v_{iy}, and the acceleration due to gravity g.

(b) Use a constant acceleration formula to find the initial vertical velocity v_{iy} required for the ball to reach the maximum height of the trajectory 4.000 m from the ground.

(c) Use a constant acceleration formula to find the time from Tegan releasing the ball until it reaches its maximum height 4.000 m from the ground.

(d) Calculate the time for the ball to fall from the maximum height 4.000 m to the height of the ring located 3.050 m from the ground.

(e) Calculate the total time for the ball to travel up to 4.000 m and fall back to 3.050 m.

(f) Calculate the initial horizontal velocity v_{ix} required to make the ball follow the trajectory shown and land in the ring.

[2 + 4 + 4 + 3 + 2 + 5 = 20 marks]

Question 8 **25 marks**

A 50 cent coin of mass 15.5 grams slides down a slope and then continues across a horizontal surface before stopping a distance L from the end of the slope (see figure 2). The coefficient of kinetic friction between the coin and both surfaces is $\mu_K = 0.60$.

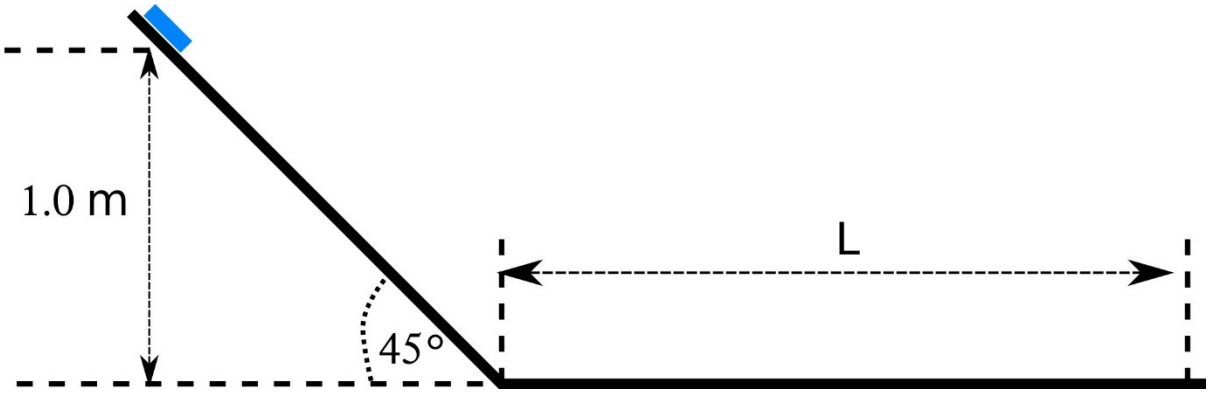


Figure 2. A 50 cent coin slides down a straight surface angled at 45 degrees to the horizontal before coming to rest a distance L beyond the end of the slope. The coefficient of kinetic friction is $\mu_K = 0.60$. The coins starts with an initial velocity of 0.50 ms^{-1}.
The initial height of the coin above the surface is 1.00 metres. The initial velocity of the coin is 0.50 ms^{-1}.

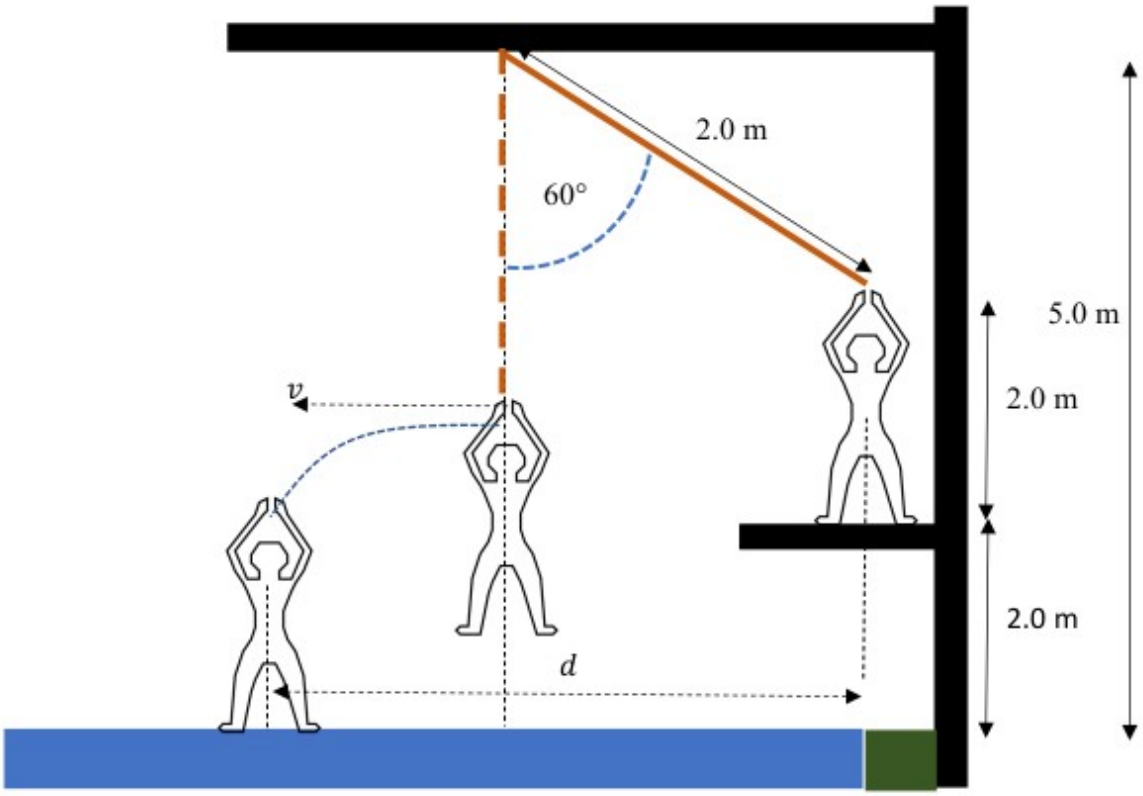
(a) Sketch a diagram in your exam booklet showing all the forces acting on the coin at the top of the slope. Include the coordinate axes you'll use to solve the problem.
(b) Write down Newton's laws of motion for the x and y coordinates you've chosen to solve the problem.

Calculate:
(c) the force due to the kinetic friction as it slides down the slope,
(d) the acceleration of the coin as it slides down the slope,
(e) the speed of the coin at the end of the slope,
(f) the kinetic energy of the coin at the bottom of the slope,
(g) the force due to kinetic friction as it slides across the horizontal surface,
(h) how much work must be done to the coin to stop its motion,
(i) how far the coin slides across the flat surface before stopping.

[5 + 2 + 2 + 4 + 4 + 2 + 2 + 2 + 2 = 25 marks]

Question 9 **30 marks**

Sophie plays with a rope attached to a tree branch hanging above a lake as shown in the figure below. Sophie has a mass of 50 kg. In all the calculations that follow, until she releases her hold on the rope, assume Sophie holds the rope at a distance 2.0 m above her feet. Neglect air resistance and friction. Use $g = 9.8\,ms^{-2}$.



(a) Sophie stands on a branch of the tree directly above the edge of the lake. She steps off the branch and swings out over the lake. Using the principle of conservation of energy, calculate Sophie's horizontal velocity v when the rope is directly vertical (i.e. at the bottom of the swing). Assume she begins her swing with an initial speed of zero.
(b) Calculate Sophie's acceleration (direction and magnitude) at this point.
(c) Draw a force diagram of Sophie and the swing and calculate the tension in the rope at this point.

Sophie releases the rope at this point and follows a projectile motion until her feet touch the lake. Assume that Sophie keeps the same body posture in the calculations that follow.

(d) Calculate the horizontal distance from the edge of the lake when she releases the rope.
(e) Calculate the time between Sophie releasing the rope and her feet hitting the water of the lake.
(f) Calculate the distance, d, from the edge of lake to where Sophie's feet hit the water.
(g) Calculate Sophie's total speed when her feet hit the water and hence her kinetic energy.

[6 + 4 + 4 + 3 + 4 + 3 + 6 = 30 marks]